

# Addressing Mechanisms and Heterogeneity in Change with Usable Tools

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<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

# Training and mentorship



The Ateneo de Manila University



Liane Peña-Alampay



University of Macau



Shu Fai Cheung



**PennState.**

The Pennsylvania State University



Sy-Miin Chow



Michael A.  
Russell

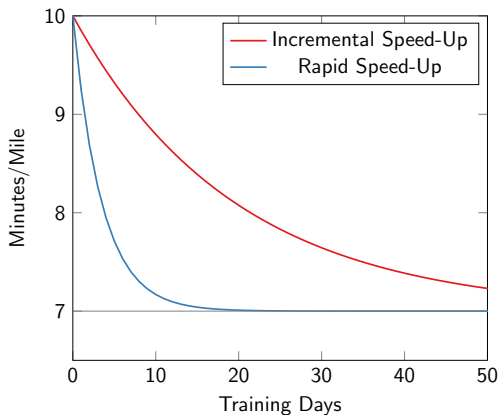
My work integrates **developmental science**, **quantitative methods**, and **dynamic modeling** to study mechanisms and heterogeneity in change across development, family life, and health—with **rigorous, usable methods**.

# An intuition for inertia



**Inertia:** how much the past carries forward into the present.

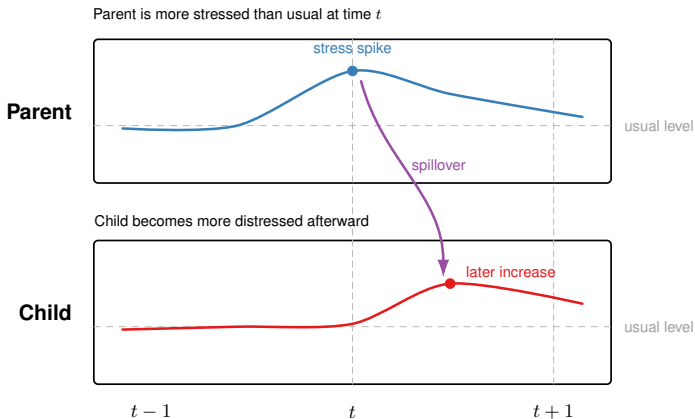
# Inertia changes trajectories



**Inertia:** The same starting point can produce very different trajectories when the **past carries forward** with different strength.

# Coupling and spillover

## Escalation / Spillover



**Coupling:** change in one process predicts later change in another.



## Inferences and Effect Sizes for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

Ivan Jacob Agaloos Pesigan<sup>1</sup>, Michael A. Russell<sup>1, 2</sup>, and Sy-Miin Chow<sup>3</sup>

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<sup>2</sup>Department of Biobehavioral Health, The Pennsylvania State University

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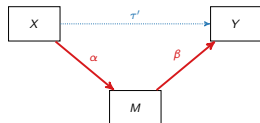
**Question:** How do mediation effects unfold across time when the process is modeled continuously rather than at one arbitrary lag?

# Why mediation has been a recurring part of my work

- ▶ **Substantive** (e.g.; Bernardo, Wang, Pesigan, & Yeung, 2017; Pesigan, Luyckx, & Alampay, 2014; Zhang, Ku, Wu, Yu, & Pesigan, 2020)
- ▶ **Methodological** (e.g.; Cheung & Pesigan, 2023b; Cheung, Pesigan, & Vong, 2023; Pesigan & Cheung, 2020, 2024; Pesigan, Sun, & Cheung, 2023).
  - ▶ <https://CRAN.R-project.org/package=semccci>
  - ▶ <https://CRAN.R-project.org/package=semblci>
  - ▶ <https://CRAN.R-project.org/package=betaDelta>
  - ▶ <https://CRAN.R-project.org/package=betaDSandwich>

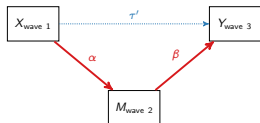
Study 1 extends that line of work into continuous time by adding **standardized effect sizes**, principled **uncertainty quantification**, and an open-source tool in cTMed.

# Why this problem matters



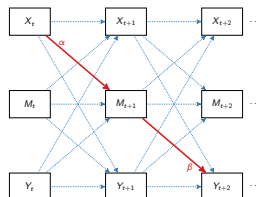
Cross-sectional

One snapshot



Three-wave

Coarse temporal ordering



Discrete-time vector  
autoregressive model

Repeated carryover across lags

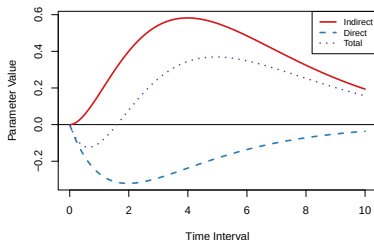
- ▶ Mediation is fundamentally a **temporal process**.
- ▶ But in discrete time, the size of the effect depends on the **chosen measurement interval**.

The **continuous-time** shift is to model the process once, then trace how direct, indirect, and total **effects unfold over any time interval**, including irregularly spaced designs.



# Novel contributions

Total, Direct, and Indirect Effects

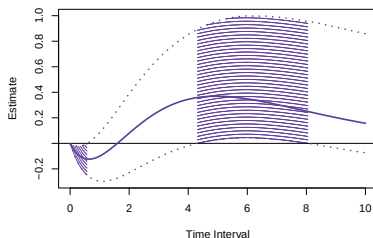


## Effect size measures

Standardized total, direct, and indirect effects

- ▶ <https://CRAN.R-project.org/package=cTMed>
- ▶ <https://CRAN.R-project.org/package=simStateSpace>
- ▶ <https://CRAN.R-project.org/package=bootStateSpace>

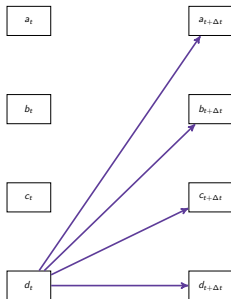
95% CI for the Total Effect (Monte Carlo Method)



## Uncertainty quantification

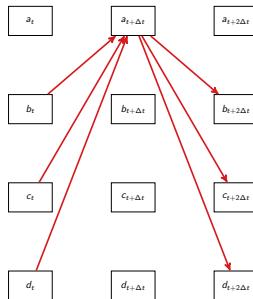
Regions of significance (delta, Monte Carlo, and parametric bootstrap)

# One direction this line of work opens up



**Total effect centrality**

Which variable matters most for shaping the rest of the system over time?



**Indirect effect centrality**

Which variable matters most as the pathway through which effects spread over time?

This framework can also be extended from single mediated pathways to **system-level summaries** of how effects spread through dynamic networks over time.

Ongoing work on person-specific affect networks: Pesigan, Russell, Chow, Fisher, & Barbara L. Fredrickson, invited submission to an *Emotion* special issue.



## Common and Unique Latent Transition Analysis (CULTA) as a Way to Examine the Trait–State Dynamics of Alcohol Intoxication

Ivan Jacob Agaloos Pesigan<sup>1</sup>, Michael A. Russell<sup>1, 2</sup>, and Sy-Miin Chow<sup>3</sup>

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**Question:** Can inertia help distinguish qualitatively different drinking patterns, beyond differences in average intoxication level?

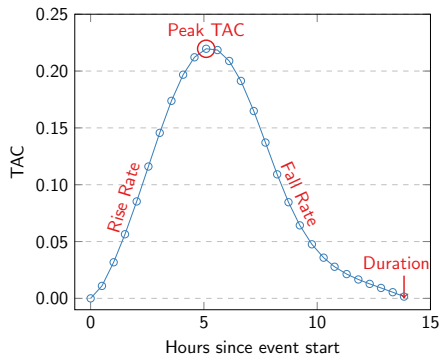
# Why this substantive problem demanded a new model

## Why Study 2 is exciting

- ▶ **Substantive grounding:** Alcohol and addiction have been a consistent part of my applied work (e.g. Yu, Pesigan, Zhang, & Wu, 2019; Yu, Wu, & Pesigan, 2015; Zamboanga, Iwamoto, Pesigan, & Tomaso, 2015; Zamboanga, Pesigan, Tomaso, Schwartz, et al., 2015; Zhang, Ku, Wu, Yu, & Pesigan, 2020; Zhang, Pesigan, Kahler, Yip, et al., 2019).
- ▶ **Substantive question:** How do risky drinking patterns persist or change across days?
- ▶ **Methodological challenge:** Transdermal alcohol concentration (TAC) features are multidimensional, partly shared, partly feature-specific, and dynamic over time.

**Substantive grounding** in addiction research,  
together with the quantitative expertise to build  
**new models** when the science demands them.

# From TAC curves to dynamic questions

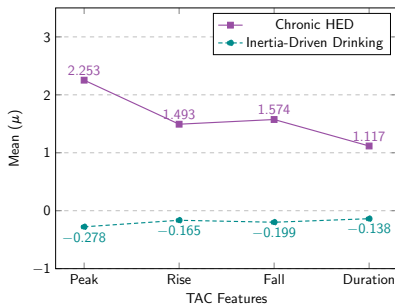


Simple summaries of the TAC features miss three things:

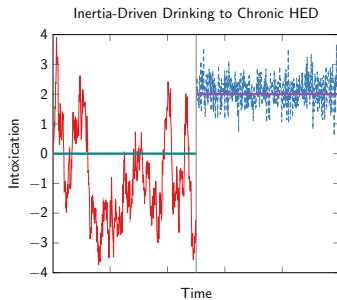
- ▶ what is **shared** across features,
- ▶ what is **feature-specific**,
- ▶ and whether intoxication shows **inertia** across days.

The question is not just **how intoxicated was this day?** but also **what kind of drinking pattern does it reflect, and how stable is that pattern over time?**

# Profiles differed in mean intoxication and inertia



Difference in mean levels



Difference in temporal persistence

- ▶ Chronic HED: Higher intoxication mean and Lower inertia
- ▶ Inertia-driven drinking: Lower intoxication mean and Higher inertia

The key point is that these profiles are **not** defined by means alone. They also differ in **how** strongly intoxication carries forward across days.

## AUD risk predicts getting stuck in chronic HED

## Start in chronic HED

No risk  $\rightarrow$  Risk  $\rightarrow$  Likely  
AUD

$$0.03 \rightarrow 0.07 \rightarrow 0.15$$

## Stay in chronic HED

No risk → Risk → Likely  
AUD

$$0.21 \rightarrow 0.30 \rightarrow 0.40$$

## Shift into chronic HED

No risk  $\rightarrow$  Risk  $\rightarrow$  Likely  
AUD

$$0.03 \rightarrow 0.06 \rightarrow 0.10$$

CULTA does not just classify drinking days; it helps identify **who is likely to get stuck in riskier dynamic patterns.**

## Study 3

Under review at *Multivariate Behavioral Research*

# From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

Ivan Jacob Agaloos Pesigan<sup>1</sup>, Michael A. Russell<sup>1,2</sup>, and Sy-Miin Chow<sup>3</sup><sup>1</sup>Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University<sup>2</sup>Department of Biobehavioral Health, The Pennsylvania State University

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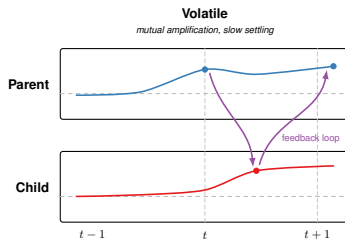
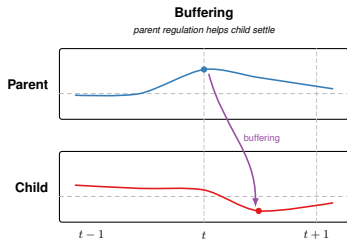
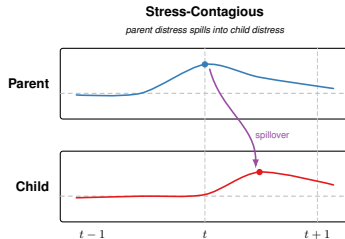
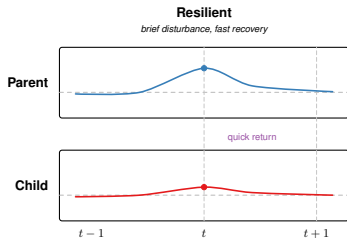
**Question:** How can we draw **population-level** conclusions from **person-specific dynamics** without averaging away the very individual differences we care about?



# Dyadic phenotypes

## Dynamic Phenotype Profiles of Dyadic Functioning

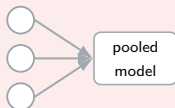
Illustrative parent-child dynamic signatures



# The trade-off MetaVAR is designed to solve

## Population inference

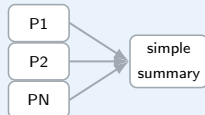
### One-step hierarchical models



- + strong for average dynamics
- partial pooling can shrink person-specific dynamics

## Idiographic fidelity

### Naive two-step workflow



- + keeps person-specific fits
- naive summaries ignore how uncertain those fits are

**MetaVAR:** estimate each person separately, then synthesize them while carrying **first-stage uncertainty** forward



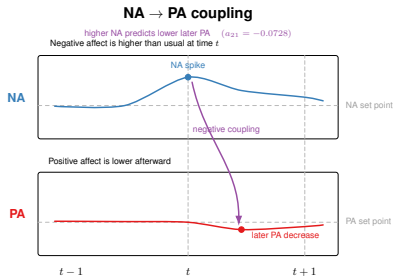
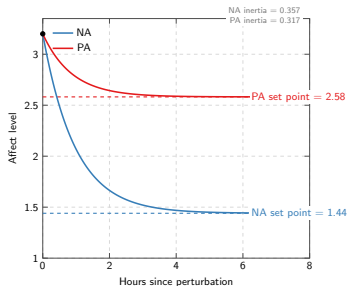
# Simulation results: MetaVAR is strongest for heterogeneity

**Coverage heatmap by parameter, design cell, and method**

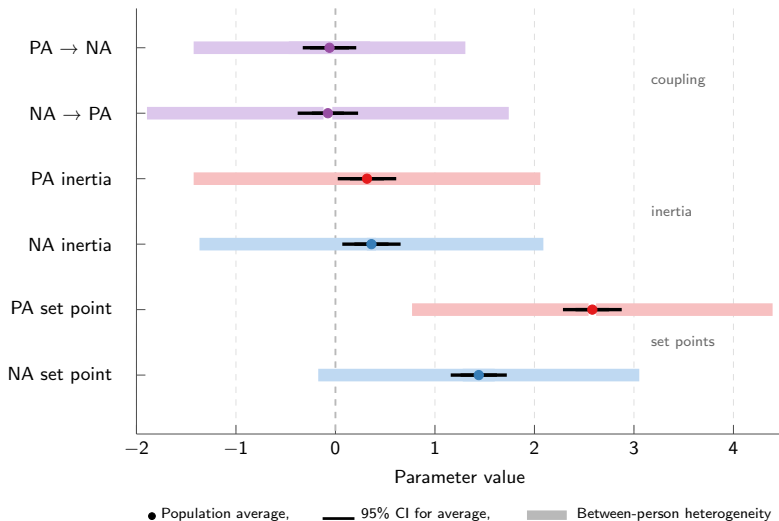
Negative values indicate undercoverage; zero indicates nominal coverage.



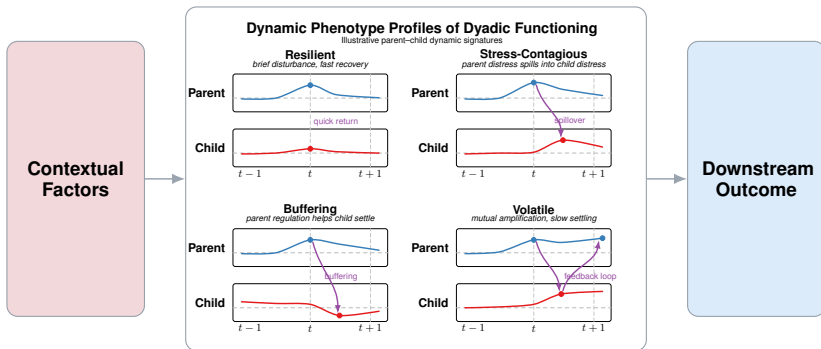
# Empirical illustration: Average affect dynamics



# Empirical illustration: Heterogeneity in affect dynamics

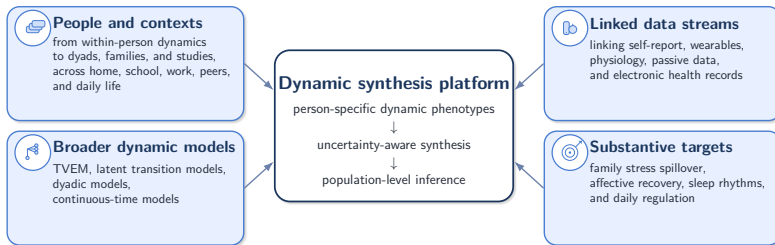


# Examining heterogeneity and downstream effects



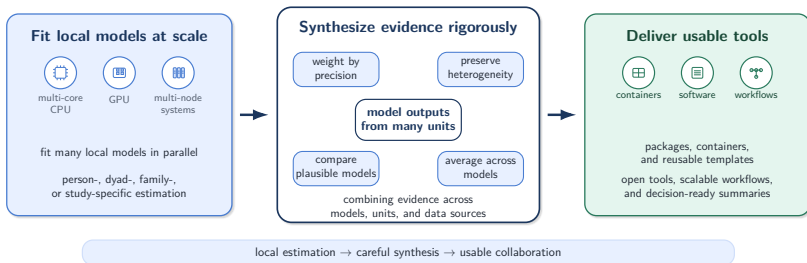
- ▶ <https://CRAN.R-project.org/package=metaDyn>
  - ▶ `meta <- MetaVARm(estimates, x = contextual, z = downstream)`

## Future directions: Toward scalable dynamic phenotypes in HDFS





## Future directions: Making dynamic synthesis usable



Matri meae carissimae, VAP.

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<https://github.com/jeksterslab/talks20260420PSUHDFSJobtalk>

# References

# References I



Bernardo, A. B., Wang, T. Y., Pesigan, I. J. A., & Yeung, S. S. (2017). Pathways from collectivist coping to life satisfaction among chinese: The roles of locus-of-hope. *Personality and Individual Differences*, 106, 253–256.  
<https://doi.org/10.1016/j.paid.2016.10.059>



Cheung, S. F., & Pesigan, I. J. A. (2023a). FINDOUT: Using either SPSS commands or graphical user interface to identify influential cases in structural equation modeling in AMOS. *Multivariate Behavioral Research*, 58(5), 964–968.  
<https://doi.org/10.1080/00273171.2022.2148089>



Cheung, S. F., & Pesigan, I. J. A. (2023b). semlbcI: An R package for forming likelihood-based confidence intervals for parameter estimates, correlations, indirect effects, and other derived parameters. *Structural Equation Modeling: A Multidisciplinary Journal*, 30(6), 985–999.  
<https://doi.org/10.1080/10705511.2023.2183860>

## References II



Cheung, S. F., Pesigan, I. J. A., & Vong, W. N. (2023). DIY bootstrapping: Getting the nonparametric bootstrap confidence interval in SPSS for any statistics or function of statistics (when this bootstrapping is appropriate).

*Behavior Research Methods*, 55(2), 474–490.

<https://doi.org/10.3758/s13428-022-01808-5>



Crocetti, E., Hale, W. W., Dimitrova, R., Abubakar, A., Gao, C.-H., & Pesigan, I. J. A. (2015). Generalized anxiety symptoms and identity processes in cross-cultural samples of adolescents from the general population. *Child & Youth Care Forum*, 44(2), 159–174.

<https://doi.org/10.1007/s10566-014-9275-9>



Deboeck, P. R., & Preacher, K. J. (2015). No need to be discrete: A method for continuous time mediation analysis.

*Structural Equation Modeling: A Multidisciplinary Journal*, 23(1), 61–75.

<https://doi.org/10.1080/10705511.2014.973960>

## References III

-  Herring, T. E., Zamboanga, B. L., Olthuis, J. V., Pesigan, I. J. A., Martin, J. L., McAfee, N. W., & Martens, M. P. (2016). Utility of the Athlete Drinking Scale for assessing drinking motives among high school athletes. *Addictive Behaviors*, 60, 18–23. <https://doi.org/10.1016/j.addbeh.2016.03.026>
-  Pesigan, I. J. A., & Cheung, S. F. (2020). SEM-based methods to form confidence intervals for indirect effect: Still applicable given nonnormality, under certain conditions. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.571928>
-  Pesigan, I. J. A., & Cheung, S. F. (2024). Monte Carlo confidence intervals for the indirect effect with missing data. *Behavior Research Methods*, 56(3), 1678–1696. <https://doi.org/10.3758/s13428-023-02114-4>

## References IV

-  Pesigan, I. J. A., Cheung, S. F., Wu, H., Chang, F., & Leung, S. O. (2026). How plausible is my model? Assessing model plausibility of structural equation models using Bayesian posterior probabilities (BPP). *Behavior Research Methods*, 58(3). <https://doi.org/10.3758/s13428-025-02921-x>
-  Pesigan, I. J. A., Luyckx, K., & Alampay, L. P. (2014). Brief report: Identity processes in Filipino late adolescents and young adults: Parental influences and mental health outcomes. *Journal of Adolescence*, 37(5), 599–604. <https://doi.org/10.1016/j.adolescence.2014.04.012>
-  Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025a). Common and unique latent transition analysis (CULTA) as a way to examine the trait-state dynamics of alcohol intoxication. *Psychology of Addictive Behaviors*, 39(8), 743–762. <https://doi.org/10.1037/adb0001106>



## References V



Pesigan, I. J. A., Russell, M. A., & Chow, S.-M. (2025b). Inferences and effect sizes for direct, indirect, and total effects in continuous-time mediation models. *Psychological Methods*. <https://doi.org/10.1037/met0000779>



Pesigan, I. J. A., Sun, R. W., & Cheung, S. F. (2023). betaDelta and betaSandwich: Confidence intervals for standardized regression coefficients in R. *Multivariate Behavioral Research*, 58(6), 1183–1186. <https://doi.org/10.1080/00273171.2023.2201277>



Russell, M. A., Richards, V. L., Turrisi, R. J., Exten, C. L., Pesigan, I. J. A., & Rodriguez, G. C. (2025). Profiles of alcohol intoxication and their associated risks in young adults' natural settings: A multilevel latent profile analysis applied to daily transdermal alcohol concentration data. *Psychology of Addictive Behaviors*, 39(2), 173–185. <https://doi.org/10.1037/adb0001022>

## References VI

-  Ryan, O., & Hamaker, E. L. (2021). Time to intervene: A continuous-time approach to network analysis and centrality. *Psychometrika*, 87(1), 214–252.  
<https://doi.org/10.1007/s11336-021-09767-0>
-  Yu, S. M., Pesigan, I. J. A., Zhang, M. X., & Wu, A. M. S. (2019). Psychometric validation of the Internet Gaming Disorder-20 Test among Chinese middle school and university students. *Journal of Behavioral Addictions*, 8(2), 295–305. <https://doi.org/10.1556/2006.8.2019.18>
-  Yu, S. M., Wu, A. M. S., & Pesigan, I. J. A. (2015). Cognitive and psychosocial health risk factors of social networking addiction. *International Journal of Mental Health and Addiction*, 14(4), 550–564.  
<https://doi.org/10.1007/s11469-015-9612-8>

## References VII



Zamboanga, B. L., Iwamoto, D. K., Pesigan, I. J. A., & Tomaso, C. C. (2015). A “player’s” game? Masculinity and drinking games participation among White and Asian American male college freshmen. *Psychology of Men & Masculinity, 16*(4), 468–473.  
<https://doi.org/10.1037/a0039101>



Zamboanga, B. L., Pesigan, I. J. A., Tomaso, C. C., Schwartz, S. J., Ham, L. S., Bersamin, M., Kim, S. Y., Cano, M. A., Castillo, L. G., Forthun, L. F., Whitbourne, S. K., & Hurley, E. A. (2015). Frequency of drinking games participation and alcohol-related problems in a multiethnic sample of college students: Do gender and ethnicity matter? *Addictive Behaviors, 41*, 112–116.  
<https://doi.org/10.1016/j.addbeh.2014.10.002>

## References VIII



Zhang, M. X., Ku, L., Wu, A. M. S., Yu, S. M., & Pesigan, I. J. A. (2020). Effects of social and outcome expectancies on hazardous drinking among Chinese university students: The mediating role of drinking motivations. *Substance Use & Misuse*, 55(1), 156–166.

<https://doi.org/10.1080/10826084.2019.1658784>



Zhang, M. X., Pesigan, I. J. A., Kahler, C. W., Yip, M. C., Yu, S., & Wu, A. M. (2019). Psychometric properties of a Chinese version of the Brief Young Adult Alcohol Consequences Questionnaire (B-YAACQ). *Addictive Behaviors*, 90, 389–394. <https://doi.org/10.1016/j.addbeh.2018.11.045>